Shunting Concentrates Path Gains and Improves Local Credit Assignment in Dendritic Networks

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Abstract

Exact gradients for conductance-based dendritic trees factorize into a synapse-local eligibility term and a single non-local compartment error; this factorization is the foundation for local credit assignment in these networks. How well a somatic broadcast can substitute for the true compartment error depends on how concentrated the conductance-stage path gains are across the tree. Across inhibited noise-resilience regimes, an oracle experiment—replacing the broadcast with the exact conductance-stage transported compartment errors—raises additive local learning from roughly 38–43% to 99–100%, nearly matching shunting (98–99%), strongly supporting path-gain fidelity as the dominant bottleneck rather than other architectural differences. Under the default shared rank-1 broadcast used by the headline feedforward architectures, shunting improves both gradient fidelity and end-task learning relative to additive integration in the inhibited regimes where conductance normalization matters: on MNIST, cosine alignment rises from 0.006 to 0.202, and on noise resilience the shunting advantage is largest once inhibition is present. We further analyze structured rank-K feedback as the natural extension when rank-1 broadcasts lose pathway identity. Together, these results identify conductance-stage path-gain concentration via shunting inhibition as the key biophysical mechanism that determines when simple shared local credit assignment works in conductance-based dendritic networks.

1 Introduction

Credit assignment in deep networks usually depends on backpropagation: global error transport through symmetric pathways with no obvious biological substrate. Dendritic compartments suggest an alternative. Each synapse has access to rich local state such as driving force, membrane conductance, and branch voltage, while supervisory information could arrive as a low-bandwidth broadcast from the soma [14]. The question here is when such local information is sufficient for useful learning.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1) and show that each synaptic gradient factorizes into purely local eligibility terms and a single non-local compartment error. Approximating that compartment error with a broadcast signal yields a theorem-facing local rule (3F), together with practical heuristic wrappers (4F/5F), and higher-rank broadcast extensions for tasks where different branches must be credited differently.

The central finding is mechanistic: shunting improves broadcast fidelity by concentrating conductance-stage path gains. On MNIST, conductance-stage path-gain CV stays at 0.21–0.27 under shunting versus 0.53–1.08 under additive integration, so one shared broadcast is a much better proxy for compartment-specific errors under shunting. On the noise-resilience sweep, per-soma broadcast reaches cosine alignment of about 0.37–0.47 under shunting across the inhibited regimes, while

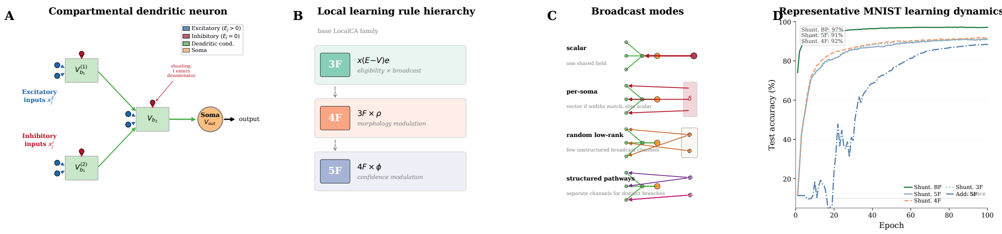

Figure 1: **Model and credit assignment. (A)** A compartmental dendritic neuron with excitatory (blue triangles) and inhibitory (red circles; shunting) synapses. Voltages propagate to the soma through learned dendritic conductances (green arrows). A somatic broadcast error e_n (dashed red) reaches all compartments; each synapse combines e_n with local quantities (presynaptic drive x_j, input resistance R_n^{tot}, and driving force $E_j - V_n$). **(B)** Compact LocalCA hierarchy: 3F is the theorem-facing local rule; 4F adds the heuristic morphology-aware factor ρ; 5F further adds the heuristic confidence-like factor ϕ. **(C)** Broadcast-mode schematic. Scalar broadcast sends one shared field to all branches; per-soma broadcast sends a vector-valued somatic error only when output and layer widths match and otherwise falls back to scalar; random low-rank broadcast adds a small number of unstructured channels; structured pathway feedback uses separate channels to preserve pathway identity when different branches should receive different teaching signals. **(D)** Representative MNIST learning dynamics from the fixed-epoch Figure 1 sweep. Shunting with the stronger local wrappers learns rapidly and remains much closer to backprop than the additive control, making the rule hierarchy concrete before the paper turns to the more precise gradient-fidelity diagnostics in Fig. 2.

additive becomes negative once inhibition is introduced even though it can still perform well at $N_I = 0$. Our question is therefore not whether local learning can match backpropagation on standard benchmarks, but what biophysical mechanism determines when low-bandwidth local credit assignment works in conductance-based dendritic networks. An oracle experiment that supplies the exact conductance-stage transported compartment errors strongly supports path-gain fidelity as the dominant bottleneck: across the inhibited noise-resilience regimes, additive local learning rises to essentially ceiling under the oracle, nearly matching shunting.

We quantify these effects with a gradient-fidelity diagnostic based on directional alignment and scale mismatch between local and backprop gradients (Fig. 2). This is a mechanism paper, not a benchmark-leadership paper: the paper's contributions are exact factorization for conductance-based dendritic trees, a mechanistic account of shunting as conductance-stage path-gain concentration, direct fidelity and oracle experiments that isolate the broadcast bottleneck, and a structured rank-K viewpoint for settings where rank-1 broadcast fails.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak conductance is fixed at 1. Each compartment voltage is therefore a conductance-weighted average. Two properties of this form matter for what follows: local sensitivities depend on both driving force $(E - V)$ and input resistance R^{tot}, and shunting inhibition corresponds to adding conductance when $E_{\text{inh}} \approx 0$.

2.1 Voltage Equation and Local Sensitivities

Consider compartment n with synaptic inputs j (activity x_j, reversal E_j, conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j, conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \qquad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \tag{1}$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \le V_n \le \max \mathcal{S}_n$ and $0 < R_n^{\mathrm{tot}} \le 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities)**.**

$$\frac{\partial V_n}{\partial g_i^{\mathrm{syn}}} = x_i R_n^{\mathrm{tot}}(E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\mathrm{den}} R_n^{\mathrm{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\mathrm{den}}} = R_n^{\mathrm{tot}}(V_i - V_n). \tag{2}$$

2.2 Shunting Inhibition as Divisive Gain Control

An inhibitory synapse with $E_{\mathrm{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\mathrm{syn}}$ and increases g_n^{tot}. Its sensitivity is $\partial V_n / \partial g_j^{\mathrm{syn}} = -x_j R_n^{\mathrm{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role.

2.3 Exact Gradients for Dendritic Trees

Let V_0 be the somatic output, $\hat{y} = W_{\mathrm{dec}} V_0$ the decoder, and $\delta_0 = W_{\mathrm{dec}}^{\top}(\partial L / \partial \hat{y})$ the somatic error.

Theorem 1 (Conductance-Stage Backpropagation on a Dendritic Tree)**.** *For a rooted dendritic tree with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, if each branch passes its pre-reactivation voltage directly to its parent, then the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} \, R_{p(n)}^{\mathrm{tot}} \, g_{n \to p(n)}^{\mathrm{den}}, \tag{3}$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the path gain

$$\alpha_n = \prod_{(i \to k) \in \mathrm{path}(n \rightsquigarrow 0)} R_k^{\mathrm{tot}} \, g_{i \to k}^{\mathrm{den}}, \qquad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \tag{4}$$

with $\alpha_0 = 1$.

Proof. Apply the chain rule on the tree-structured computation graph using Prop. 1. □

Corollary 1 (Local–Global Factorization)**.** *The exact synaptic gradient at compartment n factorizes as:*

$$\frac{\partial L}{\partial g_i^{\mathrm{syn}}} = \underbrace{x_i \, R_n^{\mathrm{tot}} \, (E_i - V_n)}_{\textit{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\textit{compartment error}}, \tag{5}$$

where the eligibility term depends only on quantities available at synapse i (presynaptic activity x_i, input resistance R_n^{tot}, driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole non-local quantity.

Under identity upward transfer, this factorization isolates the only non-local quantity as $\partial L / \partial V_n = \alpha_n \delta_0$. With a post-voltage activation $a_n = f_n(V_n)$ transmitted to the parent, the exact recursion becomes $\partial L / \partial V_n = f_n'(V_n) \, (\partial L / \partial V_{p(n)}) \, R_{p(n)}^{\mathrm{tot}} g_{n \to p(n)}^{\mathrm{den}}$, so the compartment error remains the sole non-local term but the effective path gain acquires local activation-derivative factors. Any practical local rule therefore depends entirely on how well a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4.2. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot}, which narrows the spread of local sensitivities across the tree. The theorem is stated for the pre-reactivation compartment voltage V_n produced by the conductance stage itself. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation. Our exact-error diagnostics are computed on the hooked pre-reactivation voltage, so the factorization check and compartment-error analyses target the conductance-stage quantity defined above rather than the post-reactivation activity.

3 Local Learning Rules

3.1 Broadcast Error Approximation

We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0. For a linear readout this reduces to $\delta_0 = W_{\text{dec}}^\top(\partial L/\partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top(\partial L/\partial \hat{y})$ so that the broadcast remains defined in the same core-space coordinates. We consider five broadcast modes of increasing structure:

- **Scalar:** $e_n = \bar{\delta} \cdot \mathbf{1}$, where $\bar{\delta} = \mathrm{mean}(\delta_0)$ is a single scalar broadcast to all compartments.
- **Per-soma:** $e_n = \delta_0$ when the layer dimension matches the output; layers with mismatched dimensions fall back to scalar.
- **Low-rank:** $e_n = \Gamma_K(\delta_0)$, where Γ_K uses a small number K of random broadcast channels to project the somatic error into a layer-specific vector field.
- **Local mismatch:** $e_n = (1-\alpha)\,\bar{\delta} \cdot \widetilde{m}_n + \alpha\,\bar{\delta}$, where $\widetilde{m}_n$ is the RMS-normalized, batch-centered parent–child voltage difference $P_n - V_n$ and $\alpha = 0.2$.
- **Pathway-vector:** $e_n = \alpha\,\bar{\delta} \cdot \mathbf{1} + (1-\alpha)\,\Gamma_n(\delta_0)$, where Γ_n projects the somatic error into router-inferred pathway roles, gates those roles by local pathway activity, and transports the resulting vector hierarchically through the block-structured dendritic tree.

We use per-soma broadcast by default. In our experiments, the local-mismatch variant performs much worse (Appendix A), indicating that broadcast quality is critical. In the headline feedforward architectures, however, hidden widths exceed decoder output dimension, so this default per-soma mode falls back to a shared scalar/rank-1 field (Appendix Table S13). Under shunting, that deliberately low-bandwidth broadcast is often sufficient on standard classification tasks, while the cue-routing benchmark in Sec. 4.9 is used to compare rank-1, random low-rank, and pathway-structured feedback when different branches should receive different teaching signals. Appendix Table S11 summarizes the broader set of implemented model families, training strategies, and broadcast modes used throughout the study.

3.2 Three-Factor Rule (3F)

Definition 1 (3F Update)**.** *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta\langle x_j R_n^{\text{tot}}(E_j - V_n)\, e_n\rangle_B, \qquad \Delta g_j^{\text{den}} = \eta\langle R_n^{\text{tot}}(V_j - V_n)\, e_n\rangle_B, \tag{6}$$

where $\langle\cdot\rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n. The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$.

Additive control. Rule (6) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler: $\Delta g_j^{\text{syn}} = \eta\langle x_j\, e_n\rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model.

3.3 Practical Heuristic Wrappers: 4F and 5F

4F (heuristic attenuation proxy). In the conductance-stage recursion (4), the path gain α_n attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$\rho_n = \mathrm{Cov}(\bar{V}_n, \bar{V}_0)/(\sqrt{\mathrm{Var}(\bar{V}_n)\,\mathrm{Var}(\bar{V}_0)} + \varepsilon).$$

High ρ_n up-weights branches whose voltages covary with the soma; low ρ_n down-weights branches where the broadcast is likely to be a poor proxy.

5F (confidence-weighted 4F). 5F adds a second heuristic factor

$$\phi_n = \mathrm{Var}(V_n)/(\sigma^2_{\mathrm{res}} + \varepsilon),$$

where σ^2_{res} is the residual variance of V_n after linear regression on the parent voltage. Operationally, ϕ_n is a branch-wise confidence gate: it becomes large when local voltages are strong and stable relative to unexplained residual fluctuations, and small when local fluctuations dominate. We clamp it to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived. The paper's core claim rests on path-gain concentration and the oracle experiment; the gain from ϕ_n should be read as practical denoising rather than theoretical necessity.

Proposition 2 (5F Update)**.**

$$\Delta g_j^{\mathrm{syn}} = \eta\, \rho_n \phi_n \langle x_j R_n^{\mathrm{tot}} (E_j - V_n)\, e_n \rangle_B. \tag{7}$$

A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

3.4 Structured pathway feedback and branch-role plasticity

For tasks with latent pathway structure, scalar broadcast can erase the distinction between different dendritic branches that should be credited differently. Recent experimental evidence supports the existence of vectorized instructive signals in cortical dendrites [43], motivating the structured feedback variant we introduce here. We therefore augment the local rule with a structured cue-routing variant, *PV-LocalCA*.

Hierarchical pathway-vector broadcast. The encoder/router defines a low-dimensional role basis over latent pathways. Instead of collapsing δ_0 to a scalar or per-soma vector, we project it into this role basis and gate each role by local pathway activity:

$$e_n^{\mathrm{PV}} = \sum_{k=1}^{K} q_{n,k} c_k,$$

where $c_k = \langle r_k, \delta_0 \rangle$ are role coefficients and $q_{n,k}$ are local branch-role weights. Scalar broadcast is the special case $K{=}1$, while PV-LocalCA is a structured rank-K extension rather than a separate learning principle.

Router-derived branch roles. Rather than assigning hand-coded apical/basal labels, each branch j receives a normalized role profile r_j inferred from router assignments and incoming block weights. Plasticity is then gated by branch selectivity and by a synapse-specific role-alignment factor $\langle r_i, r_j \rangle$, biasing learning toward pathway-consistent afferents without imposing a fixed branch taxonomy.

4 Experiments

4.1 Setup

We evaluate two settings: a *capacity-calibrated* regime, where matched architectures achieve strong backprop performance, and *controlled sweeps* that isolate inhibition strength, broadcast mode, and architecture. Our main datasets are MNIST, Fashion-MNIST [37], and four synthetic tasks: *context gating*, *noise resilience*, *info shunting*, and *cue integration*; formal definitions are in Appendix I. Architectures include point MLP baselines and dendritic cores with either additive integration or shunting. Main comparisons are architecture-matched additive vs. shunting cores and local vs. backprop training; DFA appears only as a supplementary baseline. Most headline MNIST, Fashion-MNIST, and context-gating results use 5F with per-soma broadcast. Because the hidden widths in these feedforward settings exceed the decoder output dimension, that default per-soma implementation reduces to a shared scalar/rank-1 broadcast in the main mechanism and competence sweeps; the higher-rank controls then use explicit low-rank, path-propagation, path-transport, or pathway-vector variants. Unless otherwise stated, dendritic layers also use the learnable post-voltage reactivation `param_tanh`; Appendix Tables S8–S9 then audit activation shape in a more controlled setting by comparing identity (`none`) against a frozen `param_tanh` under matched additive/shunting

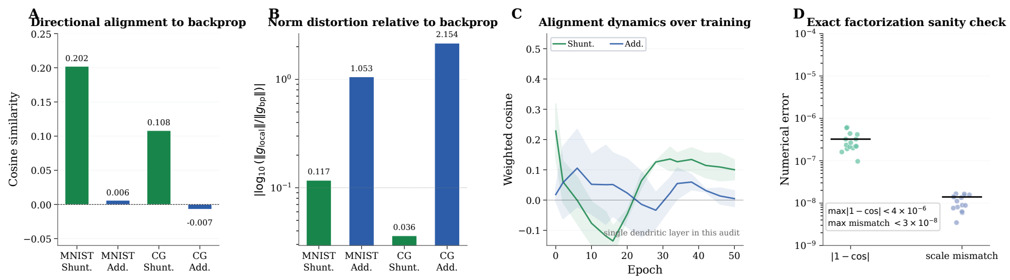

Figure 2: **Gradient-fidelity diagnostic. (A)** Weighted cosine similarity between local and backprop gradients: shunting (green) is consistently more aligned than additive (blue) on both MNIST and context gating. **(B)** Gradient scale mismatch on the same matched-weight comparisons. Shunting is not only better aligned, but also substantially closer in norm. **(C)** Per-layer cosine similarity over training epochs. Shunting proximal layers approach high positive alignment; additive layers remain near zero. **(D)** Exact-factorization sanity check across all theory-diagnostic conditions. Reconstructed gradients match autograd to numerical precision, showing that LocalCA's only approximation is in the broadcast substitute for the compartment error.

initialization. The cue-routing benchmark is used to compare rank-1, random low-rank, and pathway-structured feedback in a routed setting, and the inhibition sweeps additionally use a transported-error oracle derived from Theorem 1 as an upper bound rather than as the proposed biological rule. Headline mechanistic, oracle, competence, activation-audit, corrected routed, and CIFAR mechanism-extension results use 5 seeds per condition; a smaller set of appendix-only support sweeps still use 3 seeds, while 1–2 seed screens are treated as exploratory and are either omitted from the main claims or explicitly labeled as such in the appendix.

4.2 From Exact Factorization to Credit-Signal Fidelity

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p, we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\mathrm{local}}\|/\|g_p^{\mathrm{bp}}\|)|$, aggregated by parameter count.

Across matched-weight comparisons, shunting is consistently more aligned with backprop and substantially closer in norm than the additive control: on MNIST, weighted cosine rises from 0.006 to 0.202 and scale mismatch falls from 1.053 to 0.117; on context gating, cosine rises from -0.007 to 0.108 and scale mismatch falls from 2.154 to 0.036. Figure 2C shows that this advantage appears throughout training, especially in proximal layers and dendritic conductances. Figure 2D confirms that the theory is operational in code: reconstructing gradients from Corollary 1 using exact compartment errors matches autograd to numerical precision (weighted cosine 1.000001 ± 0.000001, weighted scale mismatch $< 4 \times 10^{-8}$). The only approximation made by LocalCA is therefore the broadcast substitute for $\partial L/\partial V_n$.

4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

Figure 3 provides the paper's central empirical chain: shunting concentrates conductance-stage path gains, which improves direct compartment-error fidelity, which in turn makes low-bandwidth local learning effective; the transported-error oracle then isolates how much of the remaining gap is due to broadcast quality rather than other architectural differences.

Panel A measures the coefficient of variation of the exact conductance-stage path gains α_n across compartments. On MNIST, shunting keeps this distribution tightly concentrated across inhibition levels (CV about 0.21–0.27), whereas the additive control remains much broader (CV about 0.53–1.08). Panel B then measures the sole non-local term from Corollary 1, $\partial L/\partial V_n$, directly. On the noise-resilience sweep, per-soma broadcast tracks the exact compartment error much better under shunting than under additive integration once inhibition is present: shunting stays around 0.40–0.47 across the inhibited settings, while additive becomes near-zero or negative even though it remains comparatively well aligned at $N_I{=}0$. The transported-error oracle obtained by recursively applying

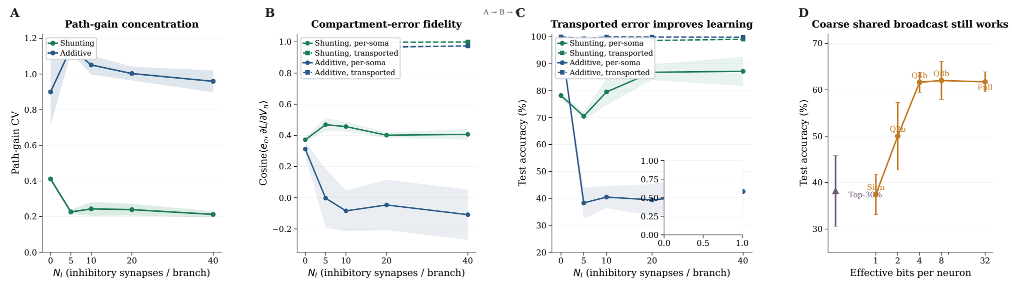

Figure 3: **Mechanistic evidence.** **(A)** Conductance-stage path-gain dispersion on MNIST, measured as the coefficient of variation of the exact tree attenuation α_n across compartments. Shunting keeps these conductance-stage path gains much more concentrated than the additive control as inhibition increases. **(B)** Direct compartment-error fidelity on noise resilience. Solid lines compare the per-soma broadcast e_n to the exact compartment error $\partial L/\partial V_n$; dashed lines show the transported-error oracle obtained by recursively applying conductance-stage path gains. Shunting substantially improves the fidelity of the simple per-soma broadcast, while the transported oracle defines the remaining approximation gap. **(C)** Learning with the transported-error oracle on noise resilience. Improving the broadcast field is enough to recover near-ceiling local learning, especially in the additive architecture where per-soma broadcast fails most severely; the CIFAR-10 inset shows the same harder-data ladder for shunting in the corrected compact E/I family. **(D)** Broadcast bandwidth. Four-bit quantization preserves full performance; only sign-only and sparse modes degrade sharply.

those conductance-stage path gains improves fidelity further in both architectures, showing that the theorem defines an operational upper bound on low-bandwidth broadcast.

Panel C trains the same local rule using that oracle signal. On noise resilience, additive learning is strongly regime-dependent under per-soma broadcast: it reaches 98.5% at $N_I{=}0$ but only about 38–43% once inhibition is added. Under transported broadcast, those inhibited additive regimes rise to 99–100%. Shunting improves from about 70–87% under per-soma to 98–99% across the same inhibited sweep. On MNIST, transported broadcast reaches about 95.4–96.7% for additive and 96.5–97.3% for shunting. The CIFAR-10 inset shows the same harder-data pattern in the corrected compact E/I family: shunting rises from 28.8% under per-soma broadcast to 47.4% under transported error, nearly matching its 49.5% backprop ceiling. These oracle results strongly support path-gain fidelity as the dominant bottleneck for LocalCA performance, while the smaller remaining additive–shunting gap under matched transported broadcast suggests an additional contribution from conductance-shaped local sensitivities.

Panel D shows that coarse broadcast can still be enough once local sensitivities are well scaled: 4-bit quantization preserves full performance (61.6% vs. 61.7%; $p > 0.9$), while sign-only (37.5%) and sparse top-30% (38.2%) degrade sharply. The corrected rank-bridge sweep in Appendix Table S7 adds one constructive intermediate control. On the same shunting noise-resilience regime, per-soma reaches only 0.333 ± 0.022, the current path-propagation proxy remains similar at 0.327 ± 0.018, random low-rank broadcast improves to 0.536 ± 0.105 at $K{=}4$, and exact path transport reaches 0.675 ± 0.019. So higher-rank unstructured feedback helps partially, whereas the current morphology-aware propagation proxy does not yet recover much of the transported-oracle gain. Normalization-only additive controls and DFA baselines (Appendix Figs. S7, S6) improve over weaker baselines without matching the shunting conductance model. Appendix Tables S8–S9 show that this mechanism should be interpreted in the bounded `param_tanh` operating regime used in the main sweeps rather than as an activation-agnostic claim.

4.4 Capacity-Calibrated Competence

Having established the mechanism, we next ask whether the same conditions improve end-task learning in architectures that are already competent under backpropagation. In the capacity-calibrated regime, shunting dendritic cores remain reasonably competitive under local learning: on MNIST, Fashion-MNIST, and context gating, the best local configuration (5F, per-soma broadcast, local decoder) stays within about 5–6 percentage points of the matched backprop ceiling (Table S4; Fig. 4). Within the local-rule family, 5F is the strongest practical wrapper, but it remains heuristic rather than

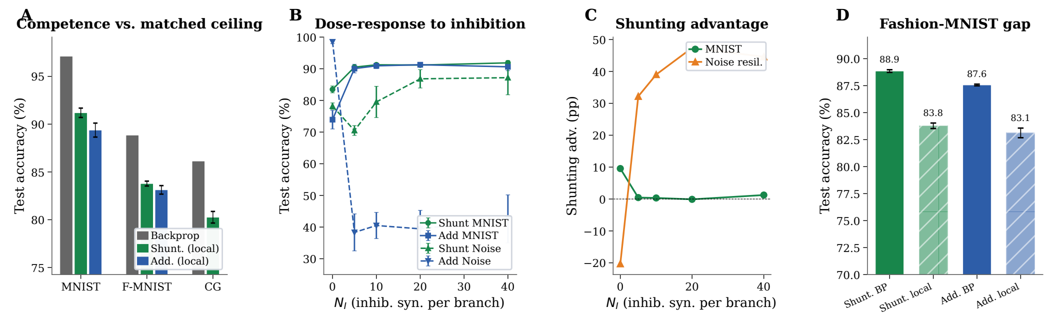

Figure 4: **Capacity-calibrated competence and regime dependence. (A)** Backprop ceiling (gray) vs. best local rule (5F per-soma) for shunting (green) and additive (blue) cores on MNIST, Fashion-MNIST, and context gating, using the refreshed 5-seed ceilings. **(B)** N_I dose-response: test accuracy vs. inhibitory synapses per branch (N_I) for shunting and additive cores on MNIST (solid) and noise resilience (dashed), updated from the activation-corrected sweep. **(C)** Shunting advantage (Δ, in percentage points) vs. N_I. **(D)** Fashion-MNIST standard-vs-local comparison from the corrected five-seed sweep, showing a smaller but still consistent shunting advantage on the cleaner vision benchmark.

theorem-derived; Appendix Tables S1, S3, and S2 summarize that division of labor across rules and broadcast choices.

4.5 Regime Dependence of the Shunting Advantage

The benefit of shunting varies across tasks. On MNIST with matched per-soma broadcast, shunting outperforms additive by about 2 percentage points; on tasks that require credit signals robust to noise, the gap is much larger: $+50.3$ pp on noise resilience (with $N_I\!=\!10$ inhibitory synapses per branch) and $+24.8$ pp on info shunting ($N_I\!=\!0$). Figure 4 shows the full dose-response: the shunting advantage grows with inhibitory conductance strength on the inhibition-sensitive tasks, whereas additive cores remain more fragile in those regimes.

4.6 Morphology Shifts the Inhibitory Operating Regime

The inhibitory regime in which shunting helps most is not fixed across dendritic trees. Figure 5 shows the corrected morphology$\times$inhibition sweep on noise resilience. The main pattern is not monotonic "more inhibition is better," but rather a morphology-dependent ridge: depth-2 trees show their largest shunting gains near $N_I\!=\!0$, whereas deeper depth-3 trees retain their strongest gains around $N_I\!=\!5$. This is consistent with the broader mechanistic picture that shunting helps by shaping the spread of effective compartment sensitivities, and suggests that tree geometry changes where that operating point lies.

4.7 Additional Stress Tests

Across three additional stress tests — increased dendritic depth, noisy broadcast signals, and Fashion-MNIST — the same qualitative pattern persists: shunting degrades more gracefully than additive, while the largest gains still appear on tasks that stress the credit signal most strongly. We keep the full curves in Appendix Fig. S5 because they support, rather than define, the core mechanistic claim.

4.8 Harder-Data Check: Corrected CIFAR-10 Mechanism Extension

To make sure the mechanism is not an artifact of near-saturated MNIST-style regimes, we repeated the core ladder on flattened CIFAR-10 in the stronger compact explicit-E/I family where shunting is already competitive under backpropagation. Figure 6 shows the corrected 5-seed result after fixing LocalCA's soma-error mapping for nonlinear decoders. In this family, shunting is stronger than additive under standard training ($49.5 \pm 0.7\%$ vs. $46.6 \pm 1.5\%$), and transported-error LocalCA again behaves as the clean upper bound: additive rises from $32.6 \pm 0.6\%$ under rank-1 per-soma broadcast to $42.3 \pm 1.2\%$ under transported error, while shunting rises from $28.8 \pm 0.8\%$ under per-soma to

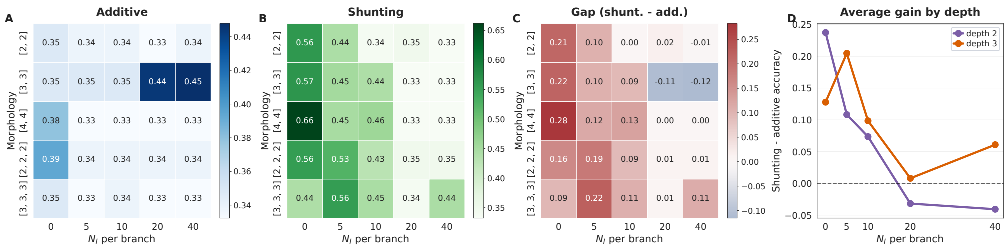

Figure 5: **Morphology shifts the inhibitory regime where shunting helps most.** **(A)** Additive LocalCA remains comparatively flat across morphologies and inhibitory synapse counts, with only isolated high-variance pockets. **(B)** Shunting LocalCA shows a clear morphology-dependent regime: the strongest performance appears at low or moderate inhibition, and the best N_I shifts with tree shape. **(C)** The shunting-minus-additive gap is largest for wider two-level trees at low inhibition and for deeper trees at modest inhibition, then weakens or vanishes at high inhibition. **(D)** Averaging by depth highlights the same crossover: depth-2 trees peak near $N_I{=}0$, whereas depth-3 trees retain their largest shunting advantage around $N_I{=}5$.

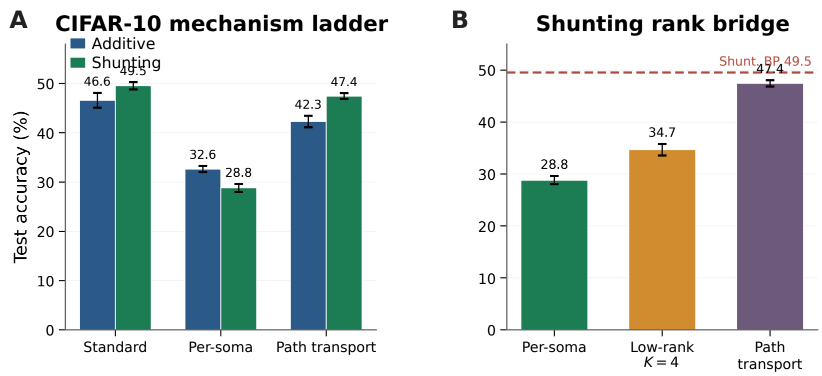

Figure 6: **Corrected CIFAR-10 mechanism extension in the stronger compact E/I family.** **(A)** Matched additive vs. shunting ladder with a nonlinear decoder and decoder-aware soma-error mapping. In this stronger family, shunting is competitive under standard training ($49.5 \pm 0.7\%$) and transported-error LocalCA nearly recovers that ceiling ($47.4 \pm 0.6\%$), while naive rank-1 per-soma remains much weaker. **(B)** Shunting rank bridge on the same family. Random low-rank feedback improves over per-soma, but exact transported error remains the strongest signal.

$34.7 \pm 1.1\%$ with random low-rank $K{=}4$ and $47.4 \pm 0.6\%$ with transported error. So the harder-data result is not that shunting always wins under naive local broadcast, but that the same broadcast-fidelity ladder still governs performance once the architecture and decoder mapping are handled fairly.

4.9 Extension: Structured Feedback When Per-Soma Broadcast Fails

The main paper focuses on low-bandwidth scalar or per-soma broadcast. We therefore include a harder cue-integration benchmark only as an exploratory extension: here different dendritic pathways should receive different teaching signals, so rank-1 feedback is insufficient by construction. Figure 7 shows that the benchmark itself is learnable, but learned-routing shunting LocalCA remains the hard case. Under the corrected activation-aware rule, default rank-1 per-soma broadcast stays unstable at roughly 0.81 ± 0.13 test accuracy; moving beyond rank-1 helps, with random low-rank feedback rising from 0.742 ± 0.022 at $K{=}1$ to 0.846 ± 0.082 at $K{=}2$, while the current tuned pathway-vector variant reaches 0.815 ± 0.126. The robust conclusion is therefore the failure mode: once rank-1 feedback collapses pathway identity, higher-rank feedback can help, but the best structured repair remains open.

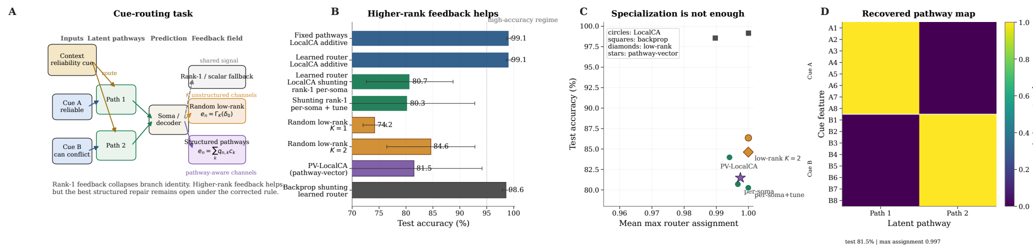

Figure 7: **Structured feedback when rank-1 broadcast fails.** **(A)** Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. **(B)** Seed-averaged diagnosis and focused tuning (5 seeds per condition). Learned-routing additive local learning remains high, whereas learned-routing shunting variants remain substantially less stable under the corrected rule; the panel compares rank-1, random low-rank, and pathway-structured variants directly. **(C)** Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. **(D)** Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate cleanly into separate latent pathways even when feedback quality, rather than router discreteness, limits performance.

5 Related Work

High-performance local learning. Recent methods such as AugLocal, CCL, and DLL have substantially raised the benchmark-performance bar for local learning on standard architectures [33, 34, 36]. Our aim is therefore not to claim the strongest benchmark accuracy among local-learning methods, but to identify the biophysical mechanism that determines when low-bandwidth credit signals become useful in conductance-based dendritic networks.

Local credit-assignment dynamics. Random feedback, DFA, DTP, DFC, equilibrium-style methods, and predictive-coding approaches solve locality through different feedback or inference dynamics [9, 10, 13, 15, 16, 25–29, 32]. EBD [35] is especially close in spirit because it studies when direct error broadcast can be made useful by shaping the broadcast field itself. Our contribution is complementary: rather than changing the broadcast objective or inference dynamics, we identify a conductance-based inductive bias—conductance-stage path-gain concentration under shunting—that makes even a simple low-bandwidth broadcast more faithful to the exact compartment error.

Dendrites for robustness and context. Dendritic trees support nonlinear computation [3, 4, 30] and have inspired learning schemes based on segregated dendrites, dendritic prediction errors, burst-dependent plasticity, and latent equilibrium [5, 11, 12, 22–24]. Outside supervised iid classification, active-dendrite architectures can also improve robustness in dynamic multi-task settings where point-MLP baselines suffer interference [44]. Our cue-routing extension asks a related question—when dendritic structure helps preserve pathway-specific computation—but uses conductance-based normalization and structured credit signals rather than sparse kWTA gating.

Neuroscience grounding. Shunting implements divisive normalization [6], though its rate effects can be subtractive in some regimes [7]; inhibitory conductance also modulates gain and signal-to-noise [31]. Recent evidence for vectorized instructive signals in cortical dendrites [43] further supports the structured-feedback extension studied here. To our knowledge, prior work has not tied exact conductance-stage path gains, direct gradient-fidelity diagnostics, and shunting inhibition into one causal account of when local dendritic learning works.

6 Discussion

We started from exact gradients for a conductance-based dendritic tree and asked when a somatic broadcast can serve as a useful proxy for compartment-specific errors. In the models studied here, the answer depends strongly on the inhibitory regime: shunting changes the conductance denominator, stabilizes effective local sensitivities, improves agreement between local and backprop gradients, and yields better local learning than an additive control, especially in noisy or inhibition-dependent settings. The morphology×inhibition map in Fig. 5 suggests that tree shape shifts this operating regime, and explicit normalization recovers only part of the benefit. This is therefore a mechanism paper, not a benchmark paper. The activation audit in the appendix shows that the mechanism lives in the bounded `param_tanh` regime used throughout the main experiments, and the oracle/rank-bridge controls sharpen the remaining open problem: finding a strictly local morphology-aware approximation that closes more of the transported-error gap. The corrected flattened-CIFAR-10 extension in Fig. 6 supports the same diagnosis in a stronger compact E/I family: once the local rule maps decoder error back to decoder-input/soma space, transported error again acts as a meaningful upper bound and shunting remains competitive on harder data.

Limitations. Our best accuracy (91.4% MNIST, 83.8% Fashion-MNIST) remains below several recent local-learning methods that operate in abstract compartments or standard activation spaces [33–36]. This reflects the constraints of conductance-based voltage space: bounded voltages, positive conductances, and a denominator-heavy computation graph. The corrected flattened-CIFAR-10 extension in Fig. 6 and Table S6 is a harder stress test rather than a claim of benchmark leadership. There shunting is competitive or better than additive under standard training and under transported-error LocalCA, but naive rank-1 per-soma broadcast still leaves a substantial gap, indicating that harder visual tasks demand better broadcast quality and likely richer front ends than flattened inputs alone. More broadly, naive local broadcasts remain much weaker than per-soma on MNIST, structured routed settings remain open, per-soma broadcast is still neuron-global, depth remains challenging, shunting is not gain-invariant in practice, and the FA and weight-statistics appendices are supportive rather than central evidence.

Broader relevance. The results are relevant to neuroscience, machine learning, and neuromorphic hardware. For neuroscience, they suggest a connection between divisive normalization and local credit fidelity. For machine learning, they show that conductance-based inductive biases can shape gradient geometry in ways that benefit local learning. For neuromorphic implementations, the update rules remain strictly local and parallelizable.

Testable predictions. Our framework predicts that (1) stronger perisomatic inhibition yields more precise synaptic plasticity; (2) GABA_A blockade selectively impairs multi-layer credit tasks; (3) the dendritic–somatic voltage correlation (ρ_n) increases during learning; and (4) learned excitatory weight distributions follow log-normal statistics whose width depends systematically on inhibitory conductance (Appendix G).

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Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	**0.912**	**0.916**
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	**0.807**	**0.789**

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

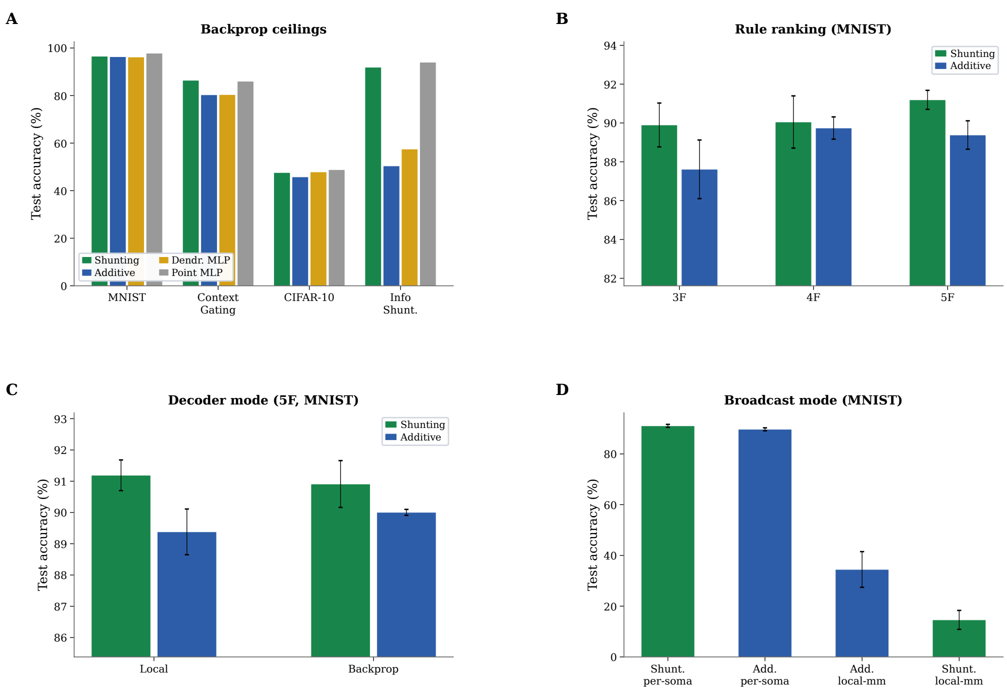

Figure S1: **Capacity calibration (supplementary).** **(A)** Phase 1 backprop ceilings across all architectures and datasets. **(B)** Rule-family ranking: 5F consistently outperforms 4F and 3F. **(C)** Decoder mode comparison (local vs. backprop decoder, 5F MNIST). **(D)** Broadcast mode comparison: per-soma is required for strong performance; local-mismatch fails.

Component	Representative effect	Interpretation
Shunting architecture	Conductance-stage path-gain CV stays at about 0.21–0.27 under shunting vs. 0.53–1.08 under additive; on inhibited noise-resilience regimes, per-soma local learning rises from roughly 0.38–0.43 (additive) to 0.70–0.87 (shunting).	This is the main architectural effect in the paper: shunting concentrates conductance-stage path gains and yields a more faithful low-bandwidth credit signal, especially once inhibition is present.
Broadcast quality	On MNIST (5F), shunting per-soma reaches 0.912 ± 0.005, while local-mismatch remains at 0.146 ± 0.046.	Correctly matching the broadcast field to the somatic error matters far more than arbitrary local mismatch heuristics.
Transported oracle	Across inhibited noise-resilience regimes, additive local learning rises from roughly 38–43% under per-soma to 99–100% under transported broadcast; shunting rises from about 70–87% to 98–99%.	The oracle is the strongest causal evidence that broadcast fidelity, not just architecture, is the dominant bottleneck.
Rule hierarchy	In the competence sweeps, MNIST improves from 0.622 (3F) and 0.628 (4F) to 0.916 (5F); context gating improves from $0.396/0.411$ to 0.789.	5F is the strongest practical rule in this paper, though its extra factor remains heuristic rather than theorem-derived.
5F stability	Tightening the ϕ clamp to $[0.5, 2.0]$ changes MNIST by only about $+0.4$ pp; varying EMA from 0.05 to 0.20 changes it by less than 0.2 pp.	The practical gain from 5F is not especially brittle to its main tuning knobs.
Structured feedback	In the corrected cue-routing reruns, learned-routing additive local learning remains near ceiling while learned-routing shunting rises from 0.742 at random rank-1 to 0.846 at random rank-2; the current tuned pathway-vector variant reaches 0.815.	Routed tasks remain the main failure mode of rank-1 feedback, but under the corrected rule the main robust conclusion is that higher rank helps; the best structured fix is not yet settled enough to anchor a headline claim.
Random low-rank channels	On the corrected shunting noise-resilience bridge, random low-rank broadcast improves from 0.423 at $K{=}1$ to 0.536 at $K{=}4$, while exact path transport reaches 0.675; on cue routing, the best corrected unstructured low-rank setting is $K{=}2$ at 0.846.	Higher-rank unstructured broadcast can close part of the rank-1 gap, but it does not recover the transported-oracle ceiling and does not by itself establish a structure-specific advantage.
Path propagation proxy	In the corrected shunting noise-resilience bridge, the current morphology-aware path-propagation proxy reaches only 0.327 ± 0.018, essentially matching per-soma (0.333 ± 0.022) and staying far below exact path transport (0.675 ± 0.019).	The present recursive attenuation proxy is too crude to recover much of the oracle gain; a strictly local approximation to transported error remains open.
Morphology regime	In the exploratory noise-resilience regime map (Fig. 5), depth-2 trees show their largest shunting gains near $N_I = 0$, whereas depth-3 trees retain their strongest gains around $N_I = 5$.	Morphology appears to shift the inhibitory operating regime in which shunting is most useful, suggesting that conductance normalization and tree geometry interact rather than contributing independently.
Activation choice	In the five-seed frozen activation audit, identity favors additive LocalCA on noise resilience ($0.662/0.647$ at $N_I{=}0/10$), whereas frozen `param_tanh` favors shunting ($0.657/0.679$). The same audit shows that frozen `param_tanh` improves shunting per-soma compartment-error cosine from 0.124 to 0.329 at $N_I{=}0$.	The post-voltage firing-rate transform is a real modeling variable, not just an implementation detail. Even a fixed saturating nonlinearity changes the operating point substantially, so the main paper's mechanistic claims should be read in the bounded `param_tanh` regime used throughout the headline experiments.

Table S2: **Which components carry the observed gains?** A compact summary of the empirical division of labor across architecture, broadcast design, and LocalCA rule variants.

A Supplementary Results

Mechanistic Support and Broadcast Controls

Rule-Family Ranking

Which Components Carry?

Broadcast Mode Comparison and Local-Mismatch Recheck

Competence, Verification, and Stress Tests

Phase 1 Capacity Ceilings

Local Competence and Regime Dependence

Ablation Results

Extended Gradient Analysis and N_I Sweep Detail

Verification and Reproducibility

Additional Stress Tests

FA/DFA Baseline Comparison

Core	Broadcast	Decoder	Test (mean±std)
Shunting	per-soma	local	0.912 ± 0.005
Shunting	per-soma	backprop	0.909 ± 0.008
Shunting	local-mismatch	local	0.146 ± 0.046
Shunting	local-mismatch	backprop	0.146 ± 0.037
Additive	per-soma	local	0.894 ± 0.007
Additive	per-soma	backprop	0.900 ± 0.001
Additive	local-mismatch	local	0.342 ± 0.058
Additive	local-mismatch	backprop	0.348 ± 0.095

Table S3: **Local-mismatch recheck (MNIST, 5F).** Per-soma is consistently strong; local-mismatch remains much weaker.

Dataset	BP ceiling	Best local (5F)	Gap
MNIST	0.971	0.914 ± 0.003	5.7%
Fashion-MNIST	0.889	0.838 ± 0.003	5.1%
Context gating	0.861	0.803 ± 0.006	5.8%

Table S4: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with per-soma broadcast on shunting dendritic cores. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix E). Error bars on local values: ± 1 s.d. across 5 seeds.

harder-data stress test no longer unfairly penalizes the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware soma mapping is used: additive LocalCA rises from 32.6% under rank-1 per-soma broadcast to 42.3% under transported error, while shunting rises from 28.8% under per-soma to 34.7% with random low-rank $K{=}4$ and 47.4% with transported error, nearly matching the 49.5% shunting backprop ceiling.

Condition	Metric	Value
Decoder: local vs backprop vs frozen (MNIST, sandbox)	test acc	0.379 vs 0.379 vs 0.176
Shunting vs additive, matched (MNIST, sandbox)	Δ test	$+0.210$
Per-soma, path on vs off (synthetic task, sandbox)	test / MI(E,I;C)	-0.003 / $+0.017$

Table S5: **Mechanistic ablations.** Results from controlled small-network sandbox experiments using minimal architectures to isolate individual factors. Sandbox runs use a small 2-layer dendritic network (branch factor [9]) trained for 30 epochs; they are not directly comparable to the headline capacity-calibrated results.

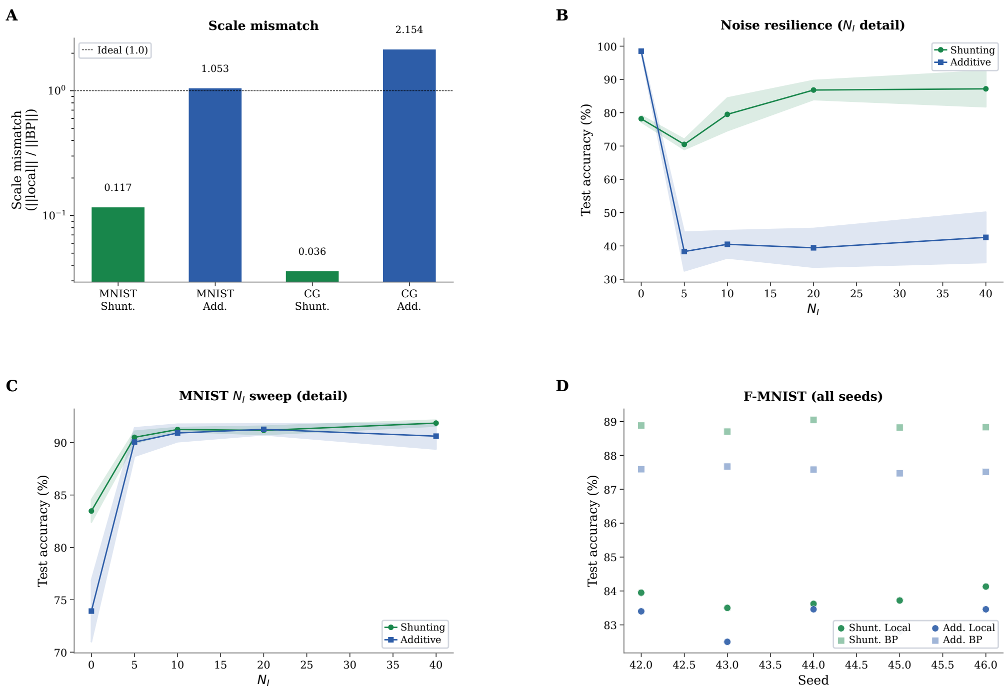

Figure S2: **Extended gradient and N_I analysis (supplementary).** **(A)** Scale mismatch bars: shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). **(B)** Noise-resilience N_I dose-response with error bands (± 1 s.d.). **(C)** MNIST N_I dose-response detail with error bands. **(D)** Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

Condition	Model	Test accuracy
standard	additive	46.6 ± 1.5
standard	shunting	$\mathbf{49.5 \pm 0.7}$
LocalCA per-soma	additive	32.6 ± 0.6
LocalCA per-soma	shunting	28.8 ± 0.8
LocalCA path transport	additive	42.3 ± 1.2
LocalCA path transport	shunting	$\mathbf{47.4 \pm 0.6}$
shunting low-rank	K=4	34.7 ± 1.1

Table S6: **Exact CIFAR-10 values for the corrected strong-family mechanism extension (5 seeds, validation-best epoch).** This table gives the exact numbers for the main-text harder-data ladder in Fig. 6. The key technical detail is that LocalCA maps output error back to decoder-input/soma space through the decoder Jacobian when the decoder is nonlinear; without that correction, the same family gave misleading near-chance results for per-soma and path transport. With the corrected rule, transported error again acts as a strong upper bound, nearly matching the stronger shunting backprop ceiling, while naive rank-1 per-soma broadcast still underperforms and random low-rank feedback closes part of that gap.

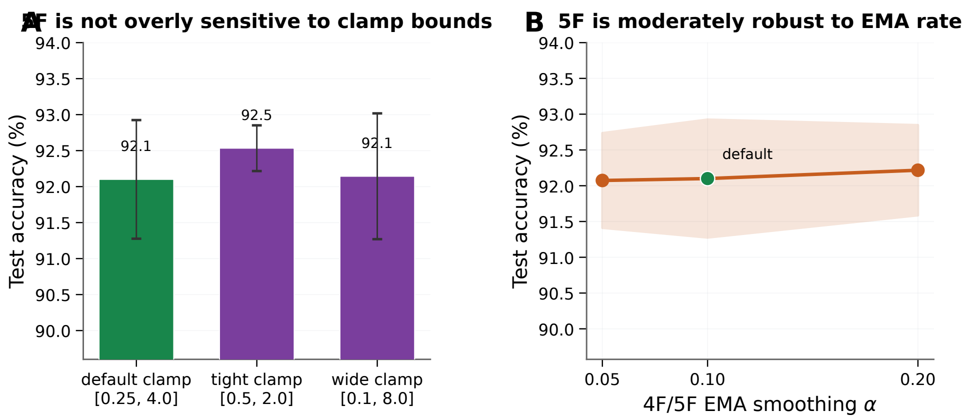

Figure S3: **5F sensitivity analysis (MNIST, shunting, per-soma broadcast).** **(A)** Sensitivity to the ϕ_n clamp interval. The default clamp $[0.25, 4.0]$ is already strong $(92.1 \pm 0.8\%)$; tightening it to $[0.5, 2.0]$ slightly improves performance $(92.5 \pm 0.3\%)$, while widening it to $[0.1, 8.0]$ is effectively neutral $(92.1 \pm 0.9\%)$. **(B)** Sensitivity to the 4F/5F EMA rate used for online statistics. Varying α from 0.05 to 0.20 changes mean test accuracy by less than 0.2 pp. Together these results suggest that the 5F gain is not driven by a brittle single hyperparameter choice, even though the factor itself remains heuristic. Errors: ± 1 s.d. across 3 seeds.

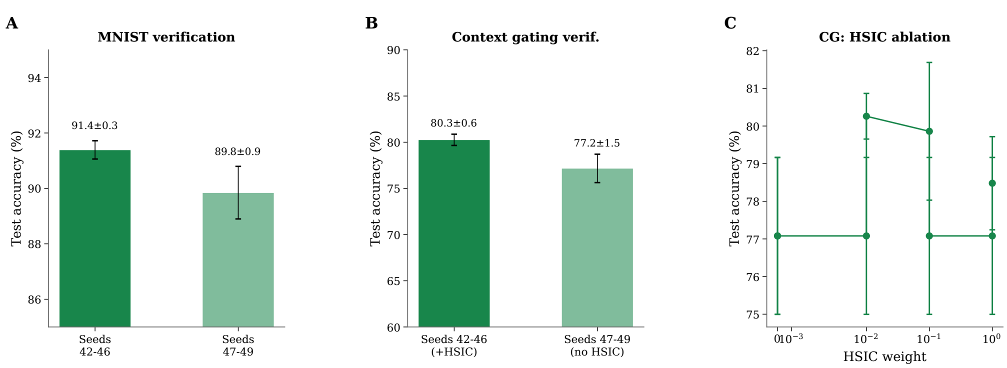

Figure S4: **Verification and reproducibility (supplementary).** **(A)** MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. **(B)** Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. **(C)** HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

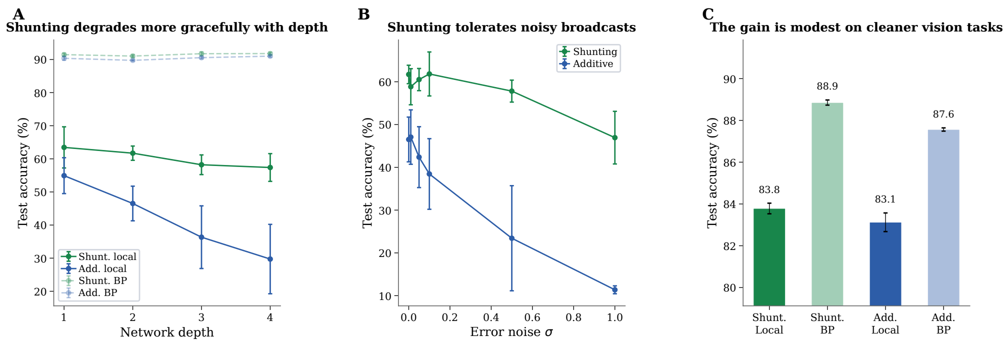

Figure S5: **Additional stress tests.** **(A)** Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. **(B)** Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. **(C)** Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

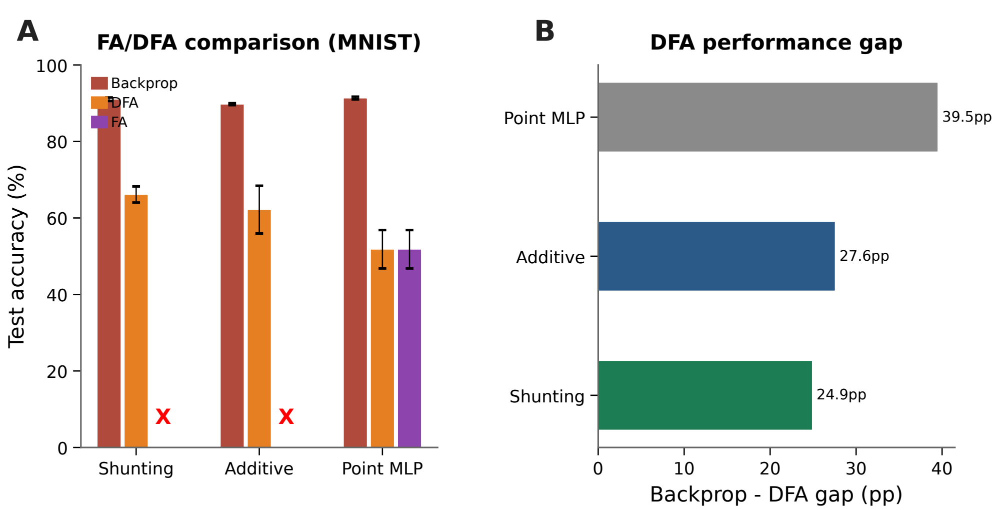

Figure S6: **Feedback alignment baselines (MNIST).** **(A)** Grouped comparison: standard backprop (gray), DFA (orange), and FA (purple). Red X marks indicate FA failure on dendritic architectures (dimensional incompatibility between random feedback matrices and block-structured dendritic layers). DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. **(B)** Backprop−DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

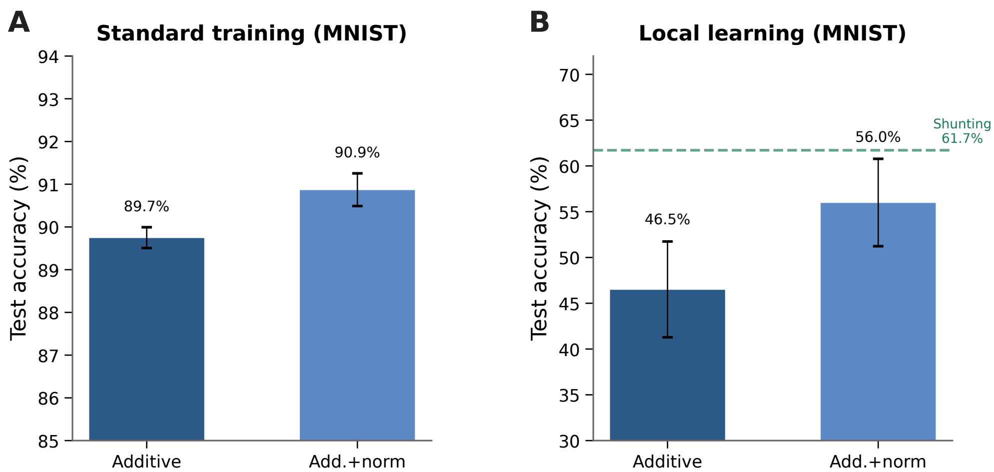

Figure S7: **Additive + normalization control (MNIST). (A)** Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). **(B)** Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (-5.7 pp) indicates that shunting provides benefits beyond simple voltage normalization.

Additive Normalization Control

Structured Feedback, Activation, and Exploratory Regimes

Morphology $\times$ Inhibition Regime Map

The main-text morphology figure shows the exploratory regime map, while the appendix keeps the summary tables and architecture details that go with it. We continue to treat it as exploratory support for the claim that morphology changes the inhibitory operating regime in which shunting is most useful; a fuller mechanistic account would require matched theory diagnostics for each cell in the grid.

Random Low-Rank Broadcast Channels

Frozen Activation-Choice Audit

B Implementation Details

Biological Plausibility Assumptions

Each synapse has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator ρ_n requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n, which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

Dendritic Structure in Code

We use the morphology notation $[b_1, b_2, \ldots, b_D]$ for rooted dendritic trees, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma; $[3, 3, 3]$ adds a third dendritic stage with 27 leaves per soma. In code, each excitatory unit is instantiated as a `DendriNet`, which is a stack of `DendriticBranchLayers`. Each branch layer contains an excitatory synapse bank, an optional inhibitory synapse bank, a

Setting	Broadcast	Test accuracy
Noise resilience, shunting, corrected rank bridge		
per-soma	rank-1 per-soma	0.333 ± 0.022
path propagation	recursive attenuation proxy	0.327 ± 0.018
low-rank	K=1	0.423 ± 0.130
low-rank	K=2	0.501 ± 0.112
low-rank	K=4	0.536 ± 0.105
low-rank	K=8	0.529 ± 0.102
path transport	exact conductance-stage transport	$\mathbf{0.675 \pm 0.019}$
Cue routing, learned shunting router, corrected rank/structure sweep		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	K=1	0.742 ± 0.022
low-rank	K=2	$\mathbf{0.846 \pm 0.082}$
low-rank	K=4	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S7: **Corrected rank bridge and rank-structure comparison.** On shunting noise resilience, random low-rank broadcast closes part of the gap between rank-1 per-soma and exact path transport, while the current path-propagation proxy does not. On cue routing, the corrected learned-router sweep shows that moving beyond rank-1 helps, but the best current unstructured low-rank setting (K=2) is slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

branch-to-parent aggregation map, and an optional post-voltage reactivation. In the main feedforward models studied here, `somatic_synapses=false`, so synaptic inputs terminate on dendritic branches rather than directly on the soma. Likewise, `input_mode=1` together with nonzero `ie_synapses_per_branch_per_layer` means that inhibition enters each dendritic branch through explicit inhibitory synapses even when we do not instantiate a separate inhibitory neuron population (`inhibitory_layer_sizes=[]`). The additive and shunting cores are therefore architecture-matched at the tree level; they differ in the branch-level voltage integration rule rather than in whether a dendritic tree exists.

Post-Voltage Reactivation

The conductance stage produces a pre-reactivation compartment voltage V_n, after which each branch layer may apply a monotone scalar nonlinearity. The codebase supports identity (`none`), fixed `tanh`/`sigmoid`/`relu`, and several parametric families including `param_tanh`, `param_tanh_only_m`, and piecewise linear–saturating variants. The main manuscript sweeps use `param_tanh`, a learnable bounded firing-rate transform of the form $(\tanh(m(V-b))+1)/2$, typically with identity transfer input activation; most headline feedforward sweeps use a linear decoder, while the appendix CIFAR extension uses a nonlinear decoder and therefore maps output error back to decoder-input/soma space through the decoder Jacobian before applying LocalCA. Unless explicitly overridden, additive and shunting cores can start from different reactivation initializations in code; activation-fairness ablations therefore match additive initialization to the shunting setting when comparing alternative reactivation choices directly. The appendix activation audit additionally fixes `param_tanh` after initialization to isolate activation shape from activation plasticity.

Units and Parameterization

Decoder Update Modes

W_{dec} maps $V_L \to \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen** ($\Delta W = 0$).

Dataset / strategy	**Core**	N_I	**Identity** (`none`)	**Frozen** `param_tanh`
MNIST / LocalCA	additive	0	0.780 ± 0.014	0.184 ± 0.049
MNIST / LocalCA	additive	10	0.907 ± 0.003	0.841 ± 0.011
MNIST / LocalCA	shunting	0	0.841 ± 0.007	0.774 ± 0.023
MNIST / LocalCA	shunting	10	0.905 ± 0.009	0.846 ± 0.015
Noise resilience / LocalCA	additive	0	0.662 ± 0.015	0.608 ± 0.022
Noise resilience / LocalCA	additive	10	0.647 ± 0.010	0.466 ± 0.034
Noise resilience / LocalCA	shunting	0	0.369 ± 0.011	0.657 ± 0.018
Noise resilience / LocalCA	shunting	10	0.368 ± 0.015	0.679 ± 0.020
MNIST / backprop	additive	0	0.925 ± 0.001	0.942 ± 0.001
MNIST / backprop	additive	10	0.921 ± 0.001	0.950 ± 0.001
MNIST / backprop	shunting	0	0.963 ± 0.0004	0.950 ± 0.001
MNIST / backprop	shunting	10	0.967 ± 0.001	0.946 ± 0.001
Noise resilience / backprop	additive	0	0.649 ± 0.015	0.995 ± 0.002
Noise resilience / backprop	additive	10	0.646 ± 0.003	0.991 ± 0.005
Noise resilience / backprop	shunting	0	0.361 ± 0.015	0.494 ± 0.015
Noise resilience / backprop	shunting	10	0.386 ± 0.024	0.885 ± 0.006

Table S8: **Frozen activation-shape audit on matched dendritic architectures (5 seeds per condition).** Identity (`none`) and a frozen bounded post-voltage reactivation `param_tanh` can lead to qualitatively different operating regimes. On the main synthetic mechanism task, identity favors additive LocalCA, whereas frozen `param_tanh` strongly favors shunting. Under standard backpropagation, the same frozen nonlinearity markedly improves additive and helps shunting most clearly at higher inhibition. These runs match additive reactivation initialization to the shunting setting and hold the activation parameters fixed, isolating activation shape from both the earlier additive/shunting initialization asymmetry in code and from activation learning itself.

Dataset	**Core**	N_I	**Path-gain CV none**	**Path-gain CV Frozen** `param_tanh`	**Per-soma cosine none**	**Per-soma cosine Frozen** `param_tanh`	**Path-transport cosine none**	**Path-transport cosine Frozen** `param_tanh`
MNIST	additive	0	0.281	0.330	0.229	0.006	0.996	0.670
MNIST	additive	10	0.310	0.863	0.210	0.074	0.963	0.222
MNIST	shunting	0	0.383	0.793	0.167	0.111	1.000	0.964
MNIST	shunting	10	0.386	1.246	0.235	0.097	1.000	0.966
Noise resilience	additive	0	0.146	0.229	0.478	-0.250	0.987	0.780
Noise resilience	additive	10	0.239	0.285	0.780	0.009	0.974	0.347
Noise resilience	shunting	0	0.299	0.316	0.124	0.329	0.315	0.678
Noise resilience	shunting	10	0.041	0.160	0.352	0.386	0.999	0.889

Table S9: **Pre-reactivation theory diagnostics for the frozen activation audit (LocalCA, 5 seeds per condition).** These diagnostics are computed on the hooked pre-reactivation compartment voltage V_n, so they track how the fixed post-voltage activation changes the operating point seen by the conductance-stage theory. The strongest qualitative pattern appears on the main synthetic mechanism task: frozen `param_tanh` improves shunting per-soma compartment-error fidelity but reduces it sharply in additive cores. We present these results as an operating-point audit rather than as a second theorem claim; the exact factorization statement in the main text remains the conductance-stage result verified in Fig. 2D.

Model and Mode Inventory

Representative Manuscript Architectures

Effective Broadcast Dimensionality in Headline Architectures

LocalCA Extension Compatibility

Seed Protocol and Claim Status

The seed depth is not uniform across every sweep, and we therefore separate headline evidence from exploratory screening. The main mechanistic gradient-fidelity sweep uses 5 seeds per condition, the corrected path-transport oracle sweep uses 5 seeds per condition, the cue-routing seed-repeat and tuning sweeps use 5 seeds per condition, the dedicated Fashion-MNIST competence sweep

Quantity	Symbol	Convention
Voltage	V	Normalized to $[-1, 1]$
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S10: Units and normalization.

Category	**Implemented modes**	**Role in this paper**
Core models	point MLP; dendritic MLP; dendritic additive; dendritic shunting; pathway-routed dendritic cores	Main comparisons use dendritic additive vs. dendritic shunting; routed cores are used for cue integration.
Training strategies	standard backprop; LocalCA; DFA; FA; transported-broadcast oracle	Main text uses standard and LocalCA; DFA/FA remain appendix baselines; path transport is an oracle upper bound.
Broadcast modes	scalar; per-soma; local mismatch; low-rank; path transport; pathway vector	Main text centers on per-soma, low-rank, and transported-error controls; scalar and local-mismatch are supporting controls, while pathway vector remains an appendix structured-feedback extension.
Recent dendritic extensions	path propagation; router-derived branch roles; pathway-vector feedback; inhibitory homeostasis	These were added to test morphology-aware credit transport and structured branch-specific plasticity beyond the original 3F/4F/5F family.

Table S11: Implemented model families and LocalCA operating modes. Only a subset is used for the main claims; the remaining modes serve as controls, ablations, or structured extensions.

uses 5 seeds per condition, the corrected flattened-CIFAR-10 mechanism extension in Table S6 also uses 5 seeds per condition, and the frozen activation audit in Tables S8–S9 uses 5 seeds per condition. A smaller set of support sweeps, including the 5F sensitivity and morphology$\times$inhibition map, use 3 seeds per condition and are described as exploratory where appropriate. By contrast, early architecture-selection or stress-test screens with only 1–2 seeds are treated as exploratory and are either omitted from the main claims or explicitly framed as pilots. For transparency, the BP ceilings in Table S4 for MNIST and context gating now come from dedicated matched-capacity 5-seed standard-training refreshes rather than the earlier 2-seed calibration pass.

Compute Resources

All experiments were run on an internal GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per seed; the flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or multi-GPU training. Reported headline metrics are aggregated over 5 seeds unless noted otherwise; this now includes the corrected flattened-CIFAR-10 mechanism extension. The remaining 3-seed results are confined to selected exploratory support sweeps, and the broader exploratory screens used to tune and diagnose the method required additional compute beyond the final reported runs.

Code, Data, and Asset Availability

MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. The present anonymous submission does not include a public code URL in order to preserve double-blind review, but the paper specifies the model family, training modes, key hyperparameters, and representative configurations, and we plan to release code and configuration files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	identity	$[128, 128]$	$[3, 3]$	N_E=40, $N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanistic sweeps in Figs. 2 and 3; dendritic reactivation is `param_tanh`.
MNIST / context-gating local competence	identity	$[128]$	$[3, 3]$	N_E=40, N_I=20	5	Best local 5F per-soma runs used in Table S4; dendritic reactivation is `param_tanh`.
Fashion-MNIST competence	identity	$[128]$	$[3, 3]$	N_E=40, N_I=20	5	Dedicated five-seed standard-vs-local competence sweep; dendritic reactivation is `param_tanh`.
Cue-routing PV-LocalCA	pathway router	$[64]$	$[2]$	N_E=10, N_I=4	5	Learned router with two pathway groups and rank-2 structured feedback; dendritic reactivation is `param_tanh`.
CIFAR-10 compact E/I mechanism extension	identity	[20] E, [20] I	$[3, 3, 3, 3]$	N_E=25, N_{EI}=25, N_{IE}=25	5	Appendix-only harder-data extension with explicit inhibitory population, decoder $[32, 16]$, and decoder-aware soma-error mapping for LocalCA.
Morphology$\times$inhibition appendix map	identity	$[128, 128]$	$[2, 2]$, $[3, 3]$, $[4, 4]$, $[2, 2, 2]$, $[3, 3, 3]$	N_E=40, $N_I \in \{0, 5, 10, 20, 40\}$	3	Exploratory appendix sweep only; dendritic reactivation is `param_tanh`.

Table S12: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false` and nonnegative conductances. The tree column gives the dendritic morphology per excitatory neuron, while the synapse counts report excitatory and inhibitory synapses per branch.

Algorithm

Algorithm 1 Local Credit Assignment

Input: Model, batch (x, y), config $\mathcal{C}$
Forward pass; loss L, output error δ^y
Somatic/core error δ_0 = decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top \delta^y$ (or $W_{\text{dec}}^\top \delta^y$ for a linear decoder)
for each layer n (reverse) **do**
    e_n = broadcast$(\delta_0, \mathcal{C})$; optionally gate by pathway roles
    Compute ρ_n, ϕ_n (EMA estimators, if enabled)
    Apply the configured local update (Eq. 6 or 7)
end for
Clip gradients; optimizer step

C Theoretical Details

Variant Taxonomy

Random-Broadcast Alignment

Proposition 3 (Supportive random-broadcast alignment)**.** *Let the broadcast matrix B_n have i.i.d. zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \alpha I$. Then the expected cosine between the resulting local gradient and the exact gradient is positive, $\mathbb{E}[\cos\angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument analogous to feedback alignment [9]. The constant c_n depends on how strongly local eligibility*

Setting	Hidden width(s)	Output dim	Effective default per-soma field
Gradient fidelity / path-gain / noise resilience	128, 128	10	Scalar / rank-1 at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Scalar / rank-1 at the dendritic hidden layer
Context gating competence	128	2	Scalar / rank-1 at the dendritic hidden layer
Cue routing	64	2	Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	decoder-input dim 20	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S13: **Effective dimensionality of the default "per-soma" broadcast in the headline architectures.** In code, per-soma uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

Extension	Meaning	Compatible with	Typical use here
Per-soma broadcast	Somatic error vector when widths match, otherwise scalar fallback	3F, 4F, 5F	Default main-text LocalCA setting; in the headline feedforward runs this is effectively rank-1
Low-rank broadcast	Small number of broadcast channels instead of a scalar or full field	3F, 4F, 5F	Intermediate control when rank-1 is too restrictive
Path transport	Recursive tree-transported error using conductance-stage path gains	3F, 4F, 5F	Oracle upper-bound broadcast
Path propagation	Approximate path-gain attenuation applied to broadcast	3F, 4F, 5F	Morphology-aware modifier for deeper trees
Pathway vector	Structured branch/pathway-specific broadcast field	3F, 4F, 5F	Used mainly in PV-LocalCA on routed tasks
Branch-role rules	Scales updates by router- or pathway-derived branch specialization	3F, 4F, 5F	Used in the structured pathway extension
Inhibitory homeostasis	Auxiliary local inhibitory balancing term	3F, 4F, 5F	Optional stabilizer in routed or shunting settings

Table S14: Compatibility of the newer LocalCA extensions with the base 3F/4F/5F family. These extensions mostly modify the error field or branch scaling rather than defining a new "6F" or "7F" rule.

factors correlate with the conductance-stage path gain in Eq. (4)*; shunting improves this correlation by narrowing the spread of intermediate sensitivities.*

We treat this argument as supportive rather than central. The main theoretical object in the paper is still the exact pre-reactivation compartment error $\partial L/\partial V_n$; under identity upward transfer this reduces to $\alpha_n \delta_0$, while with post-voltage activation the corresponding effective path gain additionally includes local $f'_n(V_n)$ factors. The main empirical question is how faithfully different broadcast fields approximate that quantity.

Rule	Factors	Cost	Best regime
3F	$x, (E-V), e$	$\mathcal{O}(1)$	Baseline
4F	3F $+\rho$	$\mathcal{O}(1)$	Improved conditioning
5F	4F $+\phi$	$\mathcal{O}(d_n)$	Strongest overall
5F+VH	5F + local voltage homeostasis	$\mathcal{O}(d_n)$	Stabilized LocalCA / warm-up
PV-LocalCA	3F + structured e_n + role alignment	$\mathcal{O}(r_n)$	Pathway discovery

Table S15: Variant taxonomy. '5F+VH' denotes the same base 5F rule with an additional strictly local voltage-centering auxiliary that can also be scheduled through multi-stage LocalCA warm-up; the codebase also supports this as a shorthand '5f_vh' alias in configuration files.

Component	Analog	Interpretation
R_n^{tot}	Input resistance	Sensitivity modulation
$(E_j - V_n)$	Synaptic driving force	Local gradient factor
Shunting	Divisive normalization	$\partial V/\partial g_I \propto -V$
ρ_n	Layer relevance	Output correlation
ϕ_n	Signal propagation	Conditional predictability
Pathway vector	Branch-specific top-down drive	Structured teaching signal
Role alignment	Pathway-selective plasticity	Branch specialization

Table S16: Biological analogs.

Biological Analogs

D Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. 4.

Depth modulation. Per-branch scaling $\rho_j = \rho_{\text{base}}/(d_j + \alpha)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}}/(\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [21].

Pathway-vector broadcast. For tasks with latent pathway structure, the broadcast becomes a low-dimensional vector over router-inferred pathway roles. This vector is gated by local pathway activity and propagated through the block-structured tree so that earlier branches receive feedback aligned with their downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

E HSIC Auxiliary Objectives

Following [17], we use kernel-based HSIC objectives. Self-decorrelation: $\mathcal{L}^{\text{self}} = B^{-2}\,\text{tr}(\mathbf{K}_Z\mathbf{H}\mathbf{K}_Z\mathbf{H})$. Target-correlation: $\mathcal{L}^{\text{target}} = -B^{-2}\,\text{tr}(\mathbf{K}_Z\mathbf{H}\mathbf{K}_Y\mathbf{H})$. Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (ρ_n, ϕ_n) use Welford's algorithm [20].

F Online Variant with Eligibility Traces

Continuous-time eligibility: $\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}}$. Update: $\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t)m_n(t)\,\text{d}t$ [18, 19].

Condition	MNIST Exc σ	MNIST Inh σ	Fashion-MNIST Exc σ	Fashion-MNIST Inh σ
BP + Shunting	0.94±0.01	0.87±0.01	0.92±0.01	0.84±0.01
BP + Additive	1.30±0.01	1.22±0.01	1.16±0.01	1.05±0.01
Local + Shunting	1.22±0.01	1.71±0.02	1.13±0.06	1.36±0.06
Local + Additive	1.69±0.16	1.57±0.07	1.43±0.16	1.11±0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E→E	$\sigma = 1.14$ (mouse V1, synapse size, n=3.9M)			
MICrONS I→E	$\sigma = 0.92$ (mouse V1, synapse size, n=3.5M)			

Table S17: **Weight distribution width.** Log-normal σ of excitatory and inhibitory synaptic weights (mean±std across 5 seeds). Biological references are included only as descriptive anchors. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.

G Weight Statistics and Biological References

As a secondary analysis, we examine whether the conductance-based update naturally induces characteristic weight statistics. Because synaptic updates scale with input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$, the voltage equation implies multiplicative weight dynamics, creating a regime analogous to exponentiated-gradient updates. Such multiplicative dynamics suggest log-normal weight distributions in trained networks [38, 39]. We compare the resulting distributions against two biological references: electrophysiological data from Song et al. [38] (rat V1 L5 pyramidal, $\sigma = 0.94$) and the MICrONS mm^3 connectome [40] (mouse V1 synapse sizes).

Model weight distributions are log-normal. We fit log-normal distributions to the active (top-K masked) excitatory synaptic weights after training on MNIST and Fashion-MNIST. Table S17 shows the fitted σ (standard deviation of $\log w$) for each condition. Shunting consistently produces narrower distributions than additive integration, and backpropagation produces narrower distributions than local learning. These two factors combine approximately additively: BP+Shunting → narrowest, Local+Additive → broadest.

Shunting is closest to the biological references considered here. BP+Shunting produces $\sigma = 0.94$ on MNIST, very close to the Song et al. reference ($\Delta = 0.008$). Local+Shunting produces $\sigma = 1.13$–1.22, close to MICrONS E→E synapse sizes ($\sigma = 1.14$; $\Delta \leq 0.08$). By contrast, Local+Additive produces $\sigma = 1.43$–1.69, broader than any biological reference considered. This pattern is consistent across datasets (MNIST and Fashion-MNIST), suggesting it reflects structural properties of the learning rule rather than task-specific features.

The match depends on inhibitory conductance. We test sensitivity to the E/I synapse ratio using a sweep over $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$ inhibitory synapses per branch (Appendix Fig. S8). Without inhibition (N_I=0), all conditions produce $\sigma > 2.0$, far from biology. With biologically constrained ratios (N_I=10–20, corresponding to E/I ratios of 4:1 to 2:1; [42]), shunting networks enter the biological range ($\sigma \approx 0.9$–1.2), while the effect of N_I on additive networks is less systematic. The weight distribution shape is also robust to dendritic depth (branch factors [9] through [3,3,3,3]; see Appendix Fig. S8A): BP+Shunting converges toward Song et al. with increasing depth, while Local+Shunting remains near MICrONS E→E across depths.

H Weight Distribution Sensitivity Analysis

MICrONS connectome analysis. We extracted synapse size distributions from the MICrONS mm^3 condensed connectome (v1181; 9.0M connections, 71.9K neurons; 89% excitatory). Splitting by pre/post cell type: E→E connections (n=3.85M) show σ_{size}=1.14 and CV=1.01; I→E (n=3.53M) show σ=0.92 and CV=1.03; E→I (n=1.11M) show σ=0.83; I→I (n=0.51M) show σ=0.85. Inhibitory connections are more frequently multi-synaptic (I→E: 37.5% multi-synapse; E→E: 11.5%), and inhibitory synapses are on average smaller (median 3532 vs. 4676 for E→E).

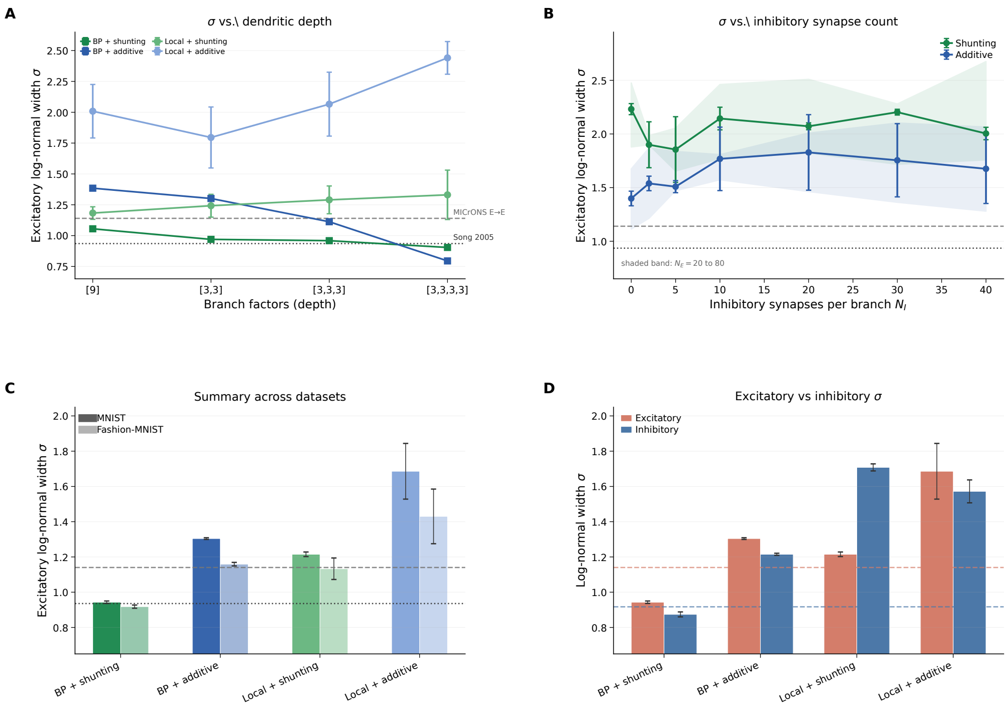

Figure S8: **Synaptic weight distribution sensitivity to architecture. (A)** Log-normal σ vs. dendritic depth (branch factors [9] to [3,3,3,3]). Both backprop and local shunting remain closer to the biological-width references than their additive counterparts across depth. **(B)** σ vs. inhibitory synapses per branch (N_I). Without inhibition (N_I=0), all conditions are overly broad; with biologically constrained N_I=10–20, shunting enters the biological range. Shaded bands show the sweep over N_E=20 to 80. **(C)** Summary of MNIST and Fashion-MNIST results with biological references. The shunting ordering is consistent across datasets. **(D)** Excitatory vs. inhibitory weight σ on MNIST with MICrONS E→E and I→E reference bands.

Sensitivity to ee_synapses. Using the E/I grid sweep ($N_E \in \{20, 40, 80\} \times N_I \in \{0, 2, 5, 10, 20, 30, 40\}$; 168 models), excitatory σ increases modestly with N_E in both architectures (+0.4–0.6 from N_E=20 to 80), but the shunting–additive ordering is preserved at all N_E values. The dominant driver of σ is N_I: going from N_I=0 to N_I=10 reduces shunting σ by approximately 1.0–1.5 units, while the effect on additive is smaller and less systematic.

Sensitivity to depth. From the depth-scaling sweep ([9], [3,3], [3,3,3], [3,3,3,3]; 80 models, 5 seeds each), BP+Shunting excitatory σ decreases with depth (1.06 $\rightarrow$ 0.90), converging toward Song et al. (0.94). Local+Shunting σ increases slightly (1.18 $\rightarrow$ 1.33) but remains in the biological range at shallow depths. BP+Additive shows the strongest depth sensitivity (1.38 $\rightarrow$ 0.79), while Local+Additive diverges with depth (2.01 $\rightarrow$ 2.44). The morphology-aware modulators (path propagation, depth/centrality weighting) have negligible effect on σ.

Mechanistic interpretation. The shunting conductance $g_n^{\text{tot}} = g_n^{\text{leak}} + \sum g^{\text{exc}} + \sum g^{\text{inh}}$ appears in the update factor $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$. As inhibitory conductance increases, g_n^{tot} grows and R_n^{tot} shrinks, compressing the multiplicative scale of weight updates. This is analogous to a learning rate that self-adjusts via the total conductance: strong synapses contribute more to g_n^{tot}, reducing their own future update magnitude. This acts as a natural form of weight-dependent regularization. The resulting equilibrium distribution is log-normal, consistent with Loewenstein et al. [41] and related multiplicative spine-dynamics analyses. Additive integration lacks this self-normalizing mechanism, producing broader distributions that fall outside the biological range.

I Synthetic Benchmark Descriptions

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. All tasks use 500-dimensional flattened inputs and 10-class outputs unless otherwise noted.

Context gating. Inputs are Gaussian mixtures over 10 classes, but the class-conditional means are linearly transformed by one of $C = 4$ context configurations. A one-hot context vector (appended to the input) indicates which configuration is active. Without the context, classes are ambiguous; with it, the optimal boundary shifts. This task tests whether credit assignment can exploit the context gating structure to modulate branch-specific representations appropriately. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are MNIST features corrupted by structured channel noise: a random 50×50 projection matrix adds correlated noise drawn from $\mathcal{N}(0, \sigma^2_{\mathrm{task}} I)$ with $\sigma_{\mathrm{task}} = 1.5$. The noise projection is fixed across training; the network must filter structured interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Info shunting. Inputs contain two additive streams: a low-frequency signal and a high-frequency distractor drawn with equal energy. A second context input (separate channel) indicates whether the relevant signal is low- or high-frequency on each trial. Correct classification requires the network to gate the distractor stream, a computation that naturally invites inhibitory modulation. Performance at $N_I{=}0$ is notably below shunting performance ($+24.8$ pp), reflecting the task's dependence on divisive normalization.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 10 classes. A binary context signal indicates which cue is reliable on each trial (cue A or cue B). The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. 4.9 to test structured pathway-vector broadcast.

J Depth Scaling and Noise Robustness

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors $[9]$ to $[3, 3, 3, 3]$): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth ($+8.5 \to +27.7$ pp). Backprop ceilings remain stable at about 90–92%. See Fig. S5A.

Noise robustness. Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive drops from 46.5% to chance at $\sigma{=}1.0$, confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. S5B.

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